

EXP 15: AI-based Traffic Prediction for 5G Networks

Objective 1: To simulate 5G network traffic data and visualize the traffic pattern.

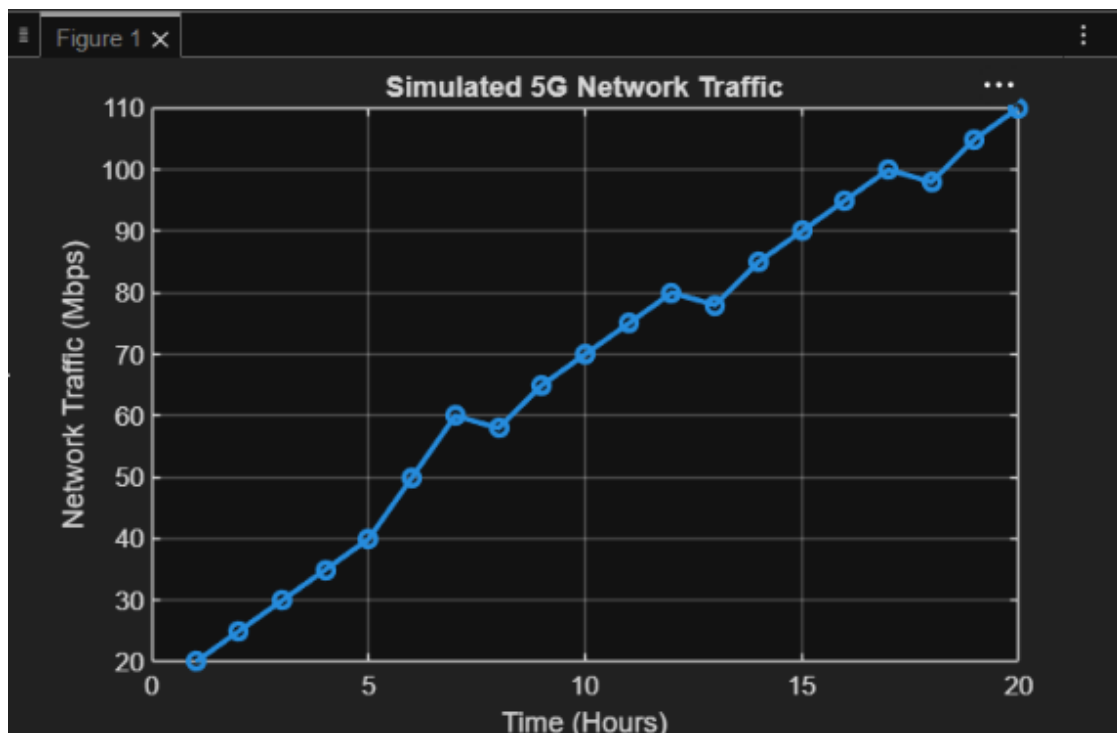
Program code:

```
clc;
clear;
close all;

% Simulated 5G Network Traffic (Mbps)
time = 1:20;
traffic = [20 25 30 35 40 50 60 58 65 70 ...
75 80 78 85 90 95 100 98 105 110];

figure;
plot(time, traffic, '-o', 'LineWidth', 2);
grid on;
xlabel('Time (Hours)');
ylabel('Network Traffic (Mbps)');
title('Simulated 5G Network Traffic');
```

Output:



Objective 2: To predict future 5G network traffic using Machine Learning (Linear Regression).

Program code:

```
clc;
clear;
close all;

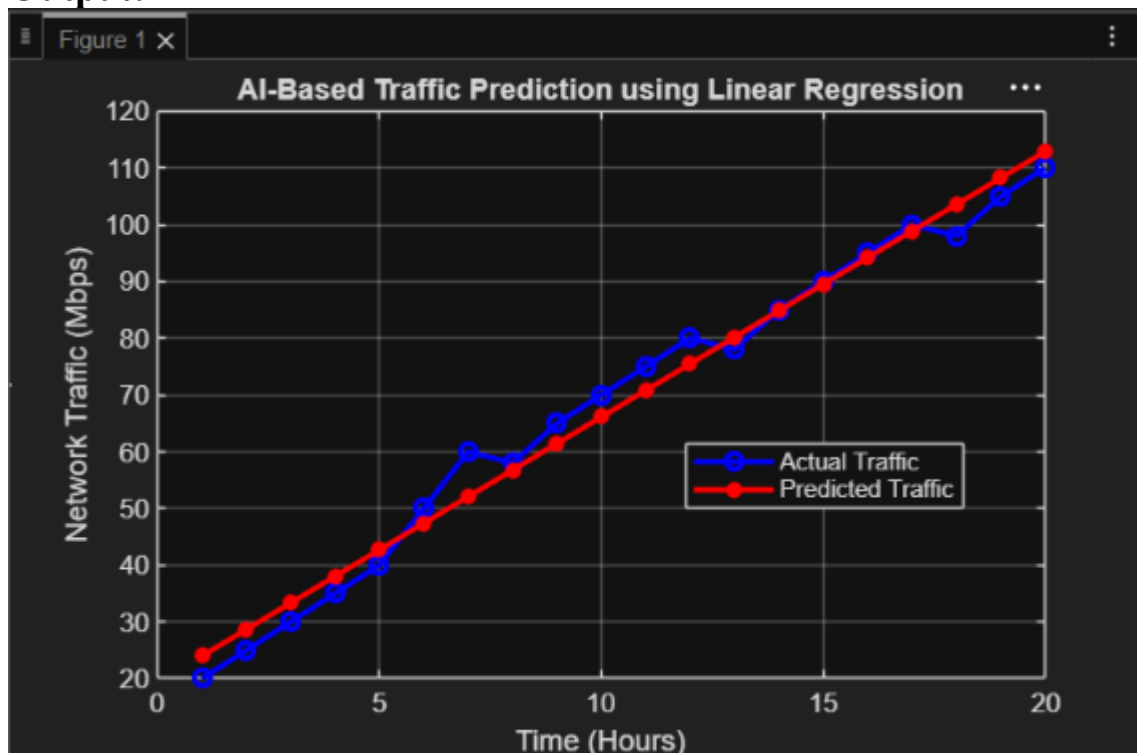
% Simulated Traffic Data
time = 1:20;
traffic = [20 25 30 35 40 50 60 58 65 70 ...
75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model
model = polyfit(time, traffic, 1);

% Predicted Traffic
predicted = polyval(model, time);

figure;
plot(time, traffic, 'bo-', 'LineWidth', 2);
hold on;
plot(time, predicted, 'r*- ', 'LineWidth', 2);
grid on;
xlabel('Time (Hours)');
ylabel('Network Traffic (Mbps)');
title('AI-Based Traffic Prediction using Linear Regression');
legend('Actual Traffic', 'Predicted Traffic', 'Location', 'best');
```

Output:



Objective 3: To evaluate the prediction accuracy by comparing actual and predicted traffic values.

Program code:

```
clc;
clear;
close all;

% Simulated Traffic Data
time = 1:20;
traffic = [20 25 30 35 40 50 60 58 65 70 ...
75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model
model = polyfit(time, traffic, 1);

% Predicted Traffic
predicted = polyval(model, time);

% Calculate Prediction Error
error = traffic - predicted;

% Calculate RMSE
RMSE = sqrt(mean(error.^2));
disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);

figure;
subplot(2,1,1);
plot(time, traffic, 'bo-', 'LineWidth', 2);
hold on;
plot(time, predicted, 'r*- ', 'LineWidth', 2);
grid on;
xlabel('Time (Hours)');
ylabel('Traffic (Mbps)');
title('Actual vs Predicted Traffic');
legend('Actual', 'Predicted', 'Location', 'best');

subplot(2,1,2);
bar(time, error);
grid on;
xlabel('Time (Hours)');
ylabel('Prediction Error (Mbps)');
title('Prediction Error');

disp('Time Actual Predicted');
disp([time' traffic' predicted]);
```

Output:

